Harmonic rank alignment and negative-scalar trace-free Ricci criticality

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Abstract

For the unrestricted functional $P(g) = \int |\operatorname{Ric}_g^\circ|^2$ on a closed oriented four-manifold, we exclude a constant-negative-scalar non-Einstein critical metric whose mixed curvature block has image in a harmonic anti-self-dual line. Neither constant amplitude nor removability of the other factor is assumed. The proof retains the full pointwise Euler–Lagrange operator, shows that the Ricci tensor must vanish at zeros of the aligned harmonic form, and tests the remaining equation at a maximum of its squared norm. A consequence is that a negative-scalar Riemannian Einstein–Maxwell metric can be P-critical only if it is Einstein. This strengthens the harmonic-alignment exclusions in the supplied manuscripts; it does not prove smooth attainment of the unrestricted infimum on the Godeaux blowup.

1 Functional, curvature block, and the statement

Let (M^4, g) be smooth, closed, connected and oriented. Set

$$E = \operatorname{Ric} - \frac{s}{4}g, \qquad P(g) = \int_M |E|^2 \, dV_g.$$

Two-forms have the form norm, with $|e^{12} + e^{34}|^2 = 2$. We compose their skew endomorphisms using g and identify the resulting self-adjoint endomorphisms with symmetric two-tensors. Write

$$\mathcal{R} = \begin{pmatrix} W^+ + sI/12 & B^* \\ B & W^- + sI/12 \end{pmatrix}, \qquad B : \Lambda^+ \longrightarrow \Lambda^-, \qquad |E|^2 = 4|B|^2.$$

The curvature contraction is $\mathscr{R}(H)_{ij} = R_{ikjl}H^{kl}$, with $R_{ikjl}g^{kl} = \operatorname{Ric}_{ij}$ and positive sectional curvature on a sphere. The rough Laplacian $L = \nabla^*\nabla$ is nonnegative.

Theorem 1.1. *Suppose s is a negative constant and $0 \neq h \in \mathcal{H}^-(M, g)$ satisfies*

$$\operatorname{im} B_x \subset \mathbb{R}h_x \qquad (x \notin Z(h)). \tag{1.1}$$

Let $\mathcal{G}_P$ be the unrestricted L^2 gradient of P. Then

$$\boxed{\|\mathcal{G}_P\|_\infty \geq \frac{|s|}{2}\|E\|_\infty.} \tag{1.2}$$

If $E \not\equiv 0$, there is a smooth nonnegative cutoff χ, supported away from $Z(h)$, such that the volume-form-preserving metrics

$$g_t(X, Y) = g(e^{-t\chi E^\sharp}X, Y) \tag{1.3}$$

satisfy $P(g_t) < P(g)$ for all sufficiently small $t > 0$. In particular, a P-critical metric satisfying (1.1) is Einstein.

The condition is exact rank alignment with a specified global harmonic form. It is not the assertion that a general P-minimizer has this structure. No lower bound for $|h|$ is required. The proof will not extend or integrate an unbounded quotient across $Z(h)$.

2 The pointwise first variation

The classical gradients of $\int |\operatorname{Ric}|^2$ and $\int s^2$ in [1, (2.5), (2.9)], combined as $P = \int |\operatorname{Ric}|^2 - \frac{1}{4}\int s^2$, give at constant scalar curvature

$$\mathcal{G}_P = LE - 2\mathscr{R}(E) + \frac{1}{2}|E|^2 g, \qquad DP_g[k] = \int_M \langle \mathcal{G}_P, k\rangle \, dV_g. \tag{2.1}$$

The trace of the full gradient at an arbitrary metric is a nonzero multiple of Δs; hence every smooth closed P-critical metric has constant scalar curvature. Scale invariance makes unit-volume criticality equivalent to unrestricted criticality.

Lemma 2.1 (Opposite-duality tensor algebra)**.** *If $E = \phi \circ \psi$ with $\phi \in \Lambda^+$ and $\psi \in \Lambda^-$, then*

$$|E|^2 = |\phi|^2|\psi|^2, \qquad (E^\sharp)^2 = \frac{|E|^2}{4} I, \tag{2.2}$$

$$\mathscr{R}(E) = (W^+\phi)\circ\psi + \phi\circ(W^-\psi) - \frac{s}{12}E + \frac{|E|^2}{4}g. \tag{2.3}$$

Here E is the trace-free Ricci tensor of the curvature used in (2.3)*.*

Proof. Opposite-duality skew endomorphisms commute and their product realizes the isometric identification $\Lambda^+ \otimes \Lambda^- \simeq S^2_0$:

$$\langle \alpha\circ\beta, \gamma\circ\delta\rangle = \langle\alpha,\gamma\rangle\langle\beta,\delta\rangle \quad (\alpha,\gamma\in\Lambda^+,\ \beta,\delta\in\Lambda^-). \tag{2.4}$$

At a point with both forms nonzero one can choose an oriented orthonormal frame with $\phi = a(e^{12}+e^{34})$, $\psi = b(e^{12}-e^{34})$. Their product is $ab\,\mathrm{diag}(-1,-1,1,1)$, proving (2.2); zero cases follow by continuity.

For completeness decompose the curvature tensor as

$$R_{ikjl} = W_{ikjl} + \frac{1}{2}(E_{ij}g_{kl} + E_{kl}g_{ij} - E_{il}g_{kj} - E_{kj}g_{il}) + \frac{s}{12}(g_{ij}g_{kl} - g_{il}g_{kj}).$$

Contraction with E^{kl} gives

$$\mathscr{R}(E) = W(E) + \frac{1}{2}|E|^2 g - E^2 - \frac{s}{12}E.$$

Under the same product identification, $W(E) = (W^+\phi)\circ\psi + \phi\circ(W^-\psi)$. This is verified by multiplication in the displayed frame for arbitrary symmetric trace-free 3×3 matrices W^+, W^-; equivariance gives the identity in all frames. Equation (2.2) proves (2.3). □

Proposition 2.2 (Local harmonic-product formula)**.** *On any open set where $E = \phi\circ\psi$ and both opposite-duality forms are smooth and harmonic, constant scalar curvature implies*

$$\boxed{\mathcal{G}_P = -\frac{s}{2}E - 2\sum_{i=1}^{4}(\nabla_{e_i}\phi)\circ(\nabla_{e_i}\psi).} \tag{2.5}$$

Writing $a = |\phi|^2$ and $b = |\psi|^2$, one has pointwise

$$\boxed{\langle \mathcal{G}_P, E\rangle = -\frac{s}{2}ab - \frac{1}{2}\langle da, db\rangle.} \tag{2.6}$$

Proof. The harmonic two-form Weitzenböck equations are

$$L\phi = 2W^+\phi - \frac{s}{3}\phi, \qquad L\psi = 2W^-\psi - \frac{s}{3}\psi.$$

Apply the product rule

$$L(\phi \circ \psi) = (L\phi) \circ \psi + \phi \circ (L\psi) - 2\sum_i (\nabla_{e_i}\phi) \circ (\nabla_{e_i}\psi).$$

Substitution of (2.3) into (2.1) cancels both Weyl terms and the scalar multiple of g, leaving (2.5). Equation (2.4) then gives

$$\sum_i \langle (\nabla_{e_i}\phi) \circ (\nabla_{e_i}\psi), \phi \circ \psi \rangle = \frac{1}{4}\langle da, db \rangle,$$

which proves (2.6). This is a pointwise calculation, so there are no discarded boundary or zero-set terms. □

3 Harmonic zeros and the maximum argument

Lemma 3.1 (Alignment at a zero)**.** *Let $h \not\equiv 0$ be a harmonic anti-self-dual two-form on a connected four-manifold, and let $B : \Lambda^+ \to \Lambda^-$ be smooth and satisfy* (1.1)*. Then $B = 0$ at every point of $Z(h)$.*

Proof. The zero set of a nonzero harmonic form has empty interior by unique continuation for the Hodge–Dirac operator [3]. Thus $\operatorname{rank} B \le 1$ everywhere, since its 2×2 minors vanish on a dense set. Suppose $h(p) = 0$ and $B(p) \neq 0$. After shrinking to a connected ball U choose a smooth local section v of Λ^+ with $Bv \neq 0$ throughout U, and put $\theta = Bv/|Bv|$. This is a smooth unit anti-self-dual form. The alignment gives $h = f\theta$ on U, including its zero set, for the smooth real function $f = \langle h, \theta \rangle$. Since $dh = 0$,

$$df \wedge \theta + f\, d\theta = 0.$$

A nonzero anti-self-dual two-form is nondegenerate. Therefore $\alpha \mapsto \alpha \wedge \theta$ is an isomorphism from one-forms to three-forms; there is a smooth one-form ω on U with $df = f\omega$. Along every path from p, this is a scalar homogeneous linear ODE. The initial value $f(p) = 0$ implies $f = 0$ on U, contradicting unique continuation of h. Hence $B(p) = 0$. □

Proof of Theorem 1.1. On $M \setminus Z(h)$ set

$$\beta_h = \frac{B^* h}{|h|^2}, \qquad \phi = -2\beta_h.$$

The mixed-block/tensor identification gives $E = \phi \circ h$. Contracted Bianchi at constant scalar curvature gives $\operatorname{div} E = 0$. For a self-dual ϕ and closed anti-self-dual h,

$$\operatorname{div}(\phi \circ h) = h(d^*\phi). \tag{3.1}$$

To check the only potentially extra term, contract $dh = 0$ with ϕ^{ik}. The two differentiated terms give the same contraction, while $\phi^{ik}\nabla_j h_{ik} = 0$ by opposite duality. Thus the derivative falling on h vanishes. This proves (3.1) with the indicated skew-endomorphism convention. As h is invertible off its zeros, $d^*\phi = 0$, and self-duality gives $d\phi = 0$. The local product formula therefore applies without any assertion of a global smooth extension of ϕ.

Assume $E \not\equiv 0$ and choose p where $|E|^2$ has its positive maximum. Lemma 3.1 and $|E|^2 = 4|B|^2$ imply $h(p) \neq 0$. Hence both $a = |\phi|^2$ and $b = |h|^2$ are positive near p and $|E|^2 = ab$. At p,

$$0 = d(ab) = b\, da + a\, db, \qquad \langle da, db \rangle = -\frac{b}{a}|da|^2 \le 0.$$

Equation (2.6) now gives

$$\langle \mathcal{G}_P, E \rangle(p) = \frac{|s|}{2}|E|^2(p) + \frac{b(p)}{2a(p)}|da|^2(p) > 0. \tag{3.2}$$

Cauchy–Schwarz at p proves (1.2). By continuity a nonnegative smooth cutoff $\chi \not\equiv 0$ can be chosen near p so that

$$DP_g[-\chi E] = -\int_M \chi \langle \mathcal{G}_P, E\rangle \, dV_g < 0.$$

The exponential in (1.3) is positive and has determinant one because $\operatorname{tr} E = 0$. It therefore gives the claimed actual metrics and first-order strict descent. For $E = 0$ all assertions are immediate. □

Remark 3.2 (The constant is sharp). For a product of two closed constant-negative-curvature surfaces with unequal Gaussian curvatures, the Ricci tensor is parallel and the two factors in its harmonic-product expression can both be chosen parallel. Then (2.5) becomes $\mathcal{G}_P = -sE/2$, and equality holds in (1.2). The theorem excludes criticality, not existence of such Einstein–Maxwell metrics.

4 Einstein–Maxwell and the Q3 interface

Corollary 4.1. *Let (g, F) be a smooth Riemannian Einstein–Maxwell solution on a closed oriented four-manifold, in the convention*

$$dF = d^*F = 0, \qquad (\mathrm{Ric} + F \circ F)_0 = 0.$$

If its scalar curvature is negative, then P-criticality of g is equivalent to g being Einstein. Every non-Einstein member admits the localized volume-preserving descent in (1.3).

Proof. The Maxwell stress tensor is divergence-free, so contracted Bianchi makes s constant. The standard four-dimensional decomposition [2] is $(F \circ F)_0 = 2F^+ \circ F^-$. Hence

$$E = (-2F^+) \circ F^-.$$

Both factors are globally smooth harmonic forms. If $F^- \equiv 0$, the metric is already Einstein. Otherwise its mixed block is aligned with F^-, and Theorem 1.1 applies. Conversely, an Einstein metric has $P = 0$ and is an absolute P-minimum. □

Corollary 4.2 (Strict rank remainder at a putative minimizer)**.** *Suppose g_* is a smooth non-Einstein P-critical metric with $s < 0$. For every $0 \neq h \in \mathcal{H}^-(M, g_*)$, define off $Z(h)$*

$$B_{\perp,h} = B - \frac{h \otimes B^* h}{|h|^2}.$$

Then $\int |B_{\perp,h}|^2 > 0$. If $b^-(M) > 0$, one further has

$$\min_{\substack{h \in \mathcal{H}^-(g_*) \\ \|h\|_2 = 1}} \int_M |B_{\perp,h}|^2 \, dV_{g_*} > 0. \tag{4.1}$$

The positive number in (4.1) *may depend on g_*.*

Proof. A vanishing integral would give (1.1) on the complement of $Z(h)$ by smoothness there, contradicting Theorem 1.1. The unit sphere of the harmonic space is compact, and convergence in it is smooth. A nonzero harmonic form has a measure-zero nodal set [3]; dominated convergence, using $|B_{\perp,h}|^2 \le |B|^2$, shows that the displayed integral is continuous in h. Its positive minimum proves (4.1). □

What this changes in the supplied proof. The supplied continuation [4] separates full flux equality from weaker rank alignment. Its global Einstein–Maxwell conclusion required boundedness of the quotient factor at zeros of the harmonic form. Theorem 1.1 removes that intermediate requirement for the negative-scalar criticality question: it uses the original smooth mixed curvature block to show $E = 0$ at those zeros and works at a positive maximum away from them. Both amplitudes are allowed to vary. The earlier integrated formula

$$DP_g[E] = -\frac{s}{2}P - \frac{1}{2}\int \langle d|\phi|^2, d|\psi|^2\rangle$$

is recovered when both factors are global, but its unsigned correlation is not used to obtain the contradiction. The full pointwise equation is the additional information.

For the fixed Godeaux blowup the supplied lower and strict upper bounds remain $16\pi^2/3 \leq P_*(M) < 8\pi^2$. No argument here proves smooth attainment of $P_*(M)$ or forces an arbitrary minimizing sequence into harmonic rank alignment. The zero-mode ALE and partial-collapse problems are not removed by this candidate-class exclusion. The present conclusions use classical curvature identities and unique continuation, not FD2 or the project's conditional gluing expansion.

References

[1] M. J. Gursky and J. A. Viaclovsky, *Rigidity and stability of Einstein metrics for quadratic curvature functionals*, arXiv:1105.4648v2 (2013). `https://arxiv.org/abs/1105.4648`. Equations (2.5), (2.9).

[2] C. LeBrun, *The Einstein–Maxwell equations, extremal Kähler metrics, and Seiberg–Witten theory*, arXiv:0803.3734v2 (2008). `https://arxiv.org/abs/0803.3734`.

[3] C. Bär, *Zero sets of solutions to semilinear elliptic systems of first order*, Invent. Math. 138 (1999), 183–202. `https://arxiv.org/abs/math/9805065`.

[4] Supplied project manuscript, *Einstein Gaps on Four-Manifolds: Subsequent Mathematical Developments*, 8 October 2026, subsection "The weaker rank-aligned system"; and the preceding note *A genuine descent direction on the harmonic-alignment branch*. These are the research context, not additional analytic hypotheses for Theorem 1.1.